

Introduction and Cross Section

Econometrics for Finance

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Stockholm School of Economics

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Outline

1 Introduction

2 Causality

3 Estimation

4 Ordinary Least Squares (OLS)

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Introduction

Welcome to the course **Econometrics for Finance**.

- My email address: morteza.aghajanzadeh@hhs.se
 - ▶ I will try to answer your questions as soon as possible.
- Three upcoming lectures (depending on class discussion, this may take more or less time)
 - ▶ Covering the basic concepts.
 - ▶ Some parts get very technical, but they are included only for formality.
 - ▶ Keep the main intuition in your mind.
 - ▶ Implement the methods in Python.
- All slides are built on courses I took during my studies.
 - ▶ I have also used slides from previous years.
 - ▶ All errors are mine.
- Office hours for technical questions:
 - ▶ Book a time slot via the [Booking Calendar](#) or [my personal website](#).
 - ▶ Send me your questions by email in advance so I can prepare.
 - ▶ If you cannot find a time slot that suits you, send me an email.

Introduction

In this course, we aim to learn how to use econometrics in finance.

At the end of this course, you should be able to:

- Understand the basic concepts of econometrics.
- Implement the econometric methods.
- Critically evaluate papers.

Important point

I will not grade your thesis, so feel free to ask me anything. Use the office hours to discuss your thesis with me.

Suggested textbooks

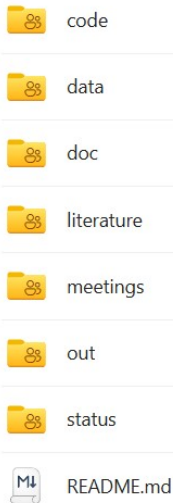
- *Introductory Econometrics for Finance* (Brooks (2019))
 - ▶ There is a free [Python Guide](#) for this book (Wichmann and Brooks (2019)).
- *Mostly Harmless Econometrics: An Empiricist's Companion* (Angrist and Pischke (2009))
- You can find more in the uploaded document.

File structure

- Literature
 - ▶ Papers
 - ▶ Reports
 - ▶ ...
- Data
 - ▶ Raw data
 - ▶ Cleaned data
- Code
 - ▶ Split your code into different parts to make it more readable.
 - ▶ Comment your code.
- Meetings
 - ▶ Per meeting summary, notes, and action items.
- Doc
 - ▶ Drafts
 - ▶ Slides
 - ▶ ...
- Output
- README.txt

File structure

Example



Research question

Research pipeline

Frequently asked questions (FAQs):

Research question

Research pipeline

Frequently asked questions (FAQs):

- 1 What is the causal relationship of interest?

Research question

Research pipeline

Frequently asked questions (FAQs):

- ① What is the causal relationship of interest?
- ② What is the ideal experiment that could be used to capture the causal effect of interest?
 - ▶ Beware of FUQs — fundamentally unidentified questions.

Research question

Research pipeline

Frequently asked questions (FAQs):

- ① What is the causal relationship of interest?
- ② What is the ideal experiment that could be used to capture the causal effect of interest?
 - ▶ Beware of FUQs — fundamentally unidentified questions.
- ③ What is your identification strategy?
- ④ What is your mode of statistical inference?

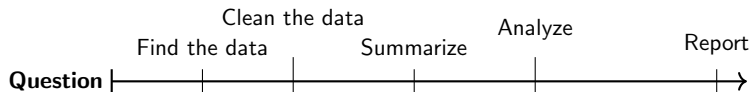
Research process

Research pipeline

Question 

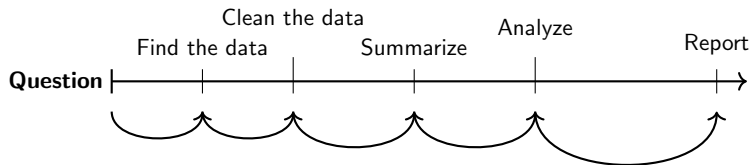
Research process

Research pipeline



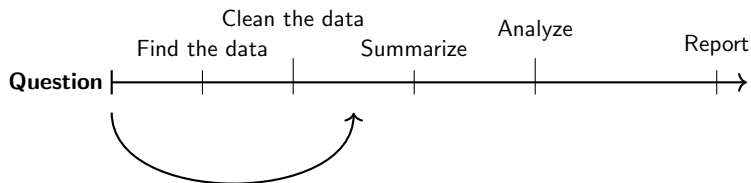
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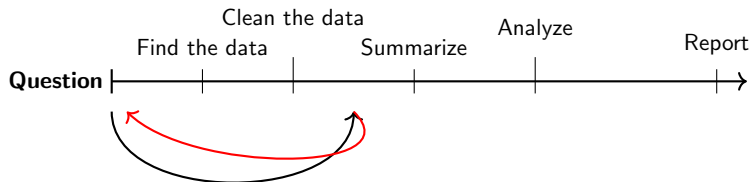
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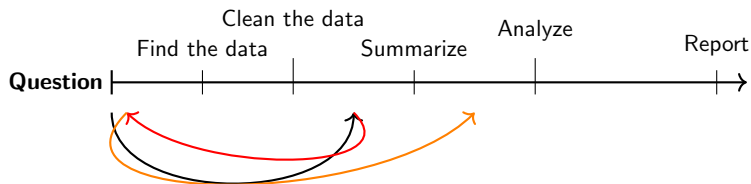
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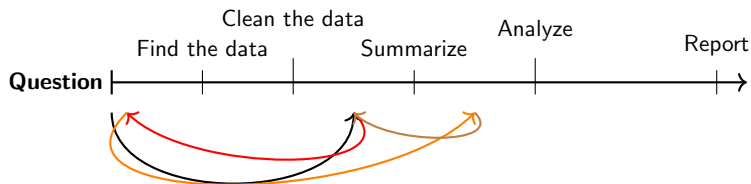
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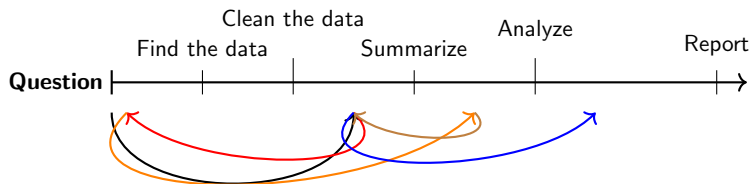
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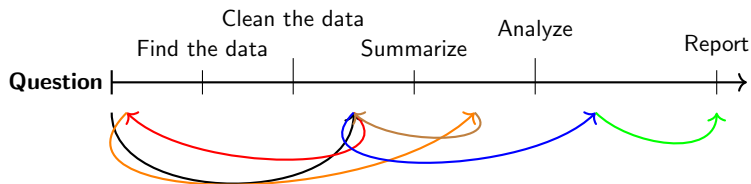
Research process

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Research process

Research pipeline



How to use AI in your research?

These are only my suggestions, not the official course policy

- You (can) use AI for everything in your research pipeline.
- But **do not** let AI do your research or your **thinking** for you.

Questions?

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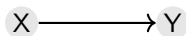
4 Ordinary Least Squares (OLS)

Causality

- What does causality mean?

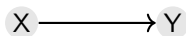
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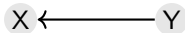
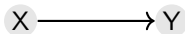
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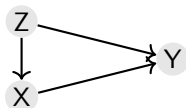
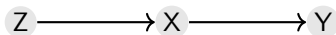
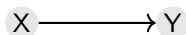
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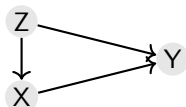
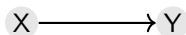
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- The most important assumption:
 - a prior ordering of the variables into causes (exogenous variables) and effects (endogenous variables).

Causality

- What does causality mean?



- The most important assumption:
 - a prior ordering of the variables into causes (exogenous variables) and effects (endogenous variables).
- Main problem:
 - We only have access to observational data.

Some common examples

- Do we really need hospitals?
- Big classes or small classes?
- Years of schooling and wages?
- Unemployment benefits and the unemployment rate?
- ...

The hospitalization question

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7,774	3.21	0.014
No hospital	90,049	3.93	0.003

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Should we conclude that going to hospital makes people less healthy?

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Should we conclude that going to hospital makes people less healthy?

- What is the problem?

Potential outcome models

Nowadays, the most common approach to causal inference is the potential outcome model.

- Let's define a binary treatment variable $D_i \in \{0, 1\}$.

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Nowadays, the most common approach to causal inference is the potential outcome model.

- Let's define a binary treatment variable $D_i \in \{0, 1\}$.
- We assume that Y_i is affected by $D_i \Rightarrow Y_i(D_i)$.
- So we have two potential outcomes for each individual:

$$\text{Potential outcome} = \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases}$$

- So what is the main problem?

Potential outcome models

- We only observe one of the potential outcomes for each individual.

$$\begin{aligned} Y_i &= \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases} \\ &= Y_i(0) + (Y_i(1) - Y_i(0))D_i \end{aligned}$$

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 - ▶ No interference between units.
 - ▶ No hidden versions of treatment.

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- Stable unit treatment value assumption (SUTVA):
 - ▶ No interference between units.
 - ▶ No hidden versions of treatment.
- We want to know the effect of the treatment on the outcome, defined as:

$$Y_i(1) - Y_i(0)$$

- ▶ And the treatment could have different effects on different individuals.

Average Treatment Effect

Definition

Average Treatment Effect (ATE) is the average of the difference between the potential outcomes for all individuals in the population.

$$ATE = \mathbb{E}[Y_i(1) - Y_i(0)]$$

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We still have the same problem:

- We cannot observe $Y_i(1)$ and $Y_i(0)$ for the same individual.

Average Treatment Effect on the Treated

Definition

Average Treatment Effect on the Treated (ATT) is the average of the difference between the potential outcomes for all individuals who receive the treatment.

$$ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

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- We still cannot measure this expectation.
- But now we are close ...
- Let's look at the observed difference in means:
 - ▶ $\mathbb{E}[Y_i \mid D_i = 1]$: average outcome for treated individuals.
 - ▶ $\mathbb{E}[Y_i \mid D_i = 0]$: average outcome for untreated individuals.

Average Treatment Effect on the Treated

$$\underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{Observed difference in means}} = \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]$$

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Average Treatment Effect on the Treated

$$\begin{aligned}\underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{Observed difference in means}} &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1] \\ &\quad + \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \text{ATT} + \underbrace{\mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0]}_{\text{Selection bias}}\end{aligned}$$

- **Selection bias:**

- ▶ The difference in average untreated outcomes $Y_i(0)$ between the treated and the untreated.
- ▶ The apples-and-oranges problem.

Average Treatment Effect on the Treated

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- **Selection bias:**

- ▶ The difference in average untreated outcomes $Y_i(0)$ between the treated and the untreated.
- ▶ The apples-and-oranges problem.

- How to eliminate the selection bias?

Assumptions

- Independence (random assignment):

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \quad \forall i$$

- Conditional independence assumption (CIA):

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i \quad \forall i$$

Key Concepts in Causality

- Endogenous and exogenous variables
- SUTVA
 - ▶ Exclusion restrictions
 - ▶ General/Partial equilibrium effects
- How to eliminate the selection bias?

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Estimation

- We want to estimate the effect of a variable x on a variable y .

$$y_i = g(X_i) + \varepsilon_i$$

- What does it mean?

Estimation

- We want to estimate the effect of a variable x on a variable y .

$$y_i = g(X_i) + \varepsilon_i$$

- What does it mean?
 - ▶ We want to estimate the function $g(\cdot)$.
 - ▶ Each estimation method has its own assumptions to estimate the function $g(\cdot)$.
 - ▶ Also, we need to make an assumption about the distribution of ε_i .

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Ordinary Least Squares (OLS)

The most common method to estimate the function $g(\cdot)$ is OLS.

Assumptions:

- $g(\cdot)$ is linear in parameters.
- $\mathbb{E}[\varepsilon_i | X_i] = 0$.

Ordinary Least Squares (OLS)

The most common method to estimate the function $g(\cdot)$ is OLS.

Assumptions:

- $g(\cdot)$ is linear in parameters.
- $\mathbb{E}[\varepsilon_i | X_i] = 0$.
- Estimation: $(\beta_{k \times 1}, X'_{i,1 \times k})$

$$Y_i = X_i' \beta + \varepsilon_i \Rightarrow \beta = \arg \min_{\beta} \mathbb{E}[(Y_i - X_i' \beta)^2]$$

$$\text{FOC: } \mathbb{E}[X_i(Y_i - X_i' \beta)] = 0 \Rightarrow \beta = \mathbb{E}[X_i X_i']^{-1} \mathbb{E}[X_i Y_i]$$

Ordinary Least Squares (OLS)

- Other formulas for β exist, but they are all equivalent (e.g., with a single regressor, $\beta = \frac{\text{Cov}(X_i, Y_i)}{\text{Var}(X_i)}$).
- We can extend it to the case of multiple variables. Stacking all n observations, the sample estimator is

$$Y = X\beta + \varepsilon \Rightarrow \hat{\beta} = (X'X)^{-1}X'Y$$

where $\beta_{k \times 1}$, $X_{n \times k} = \begin{bmatrix} X_1' \\ \vdots \\ X_n' \end{bmatrix}_{n \times k}$, and $Y_{n \times 1}$.

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$$\text{where } \beta_{k \times 1}, \quad X_{n \times k} = \begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix}_{n \times k}, \quad \text{and } Y_{n \times 1}.$$

Let's skip the technical details and focus on the intuition.

Conditional Expectation Function (CEF)

- What is the expected value of Y given X ?

$$\mathbb{E}[Y|X] = \mathbb{E}[Y_i|X_i] = m(X_i)$$

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- Linear CEF: $m(X_i) = X_i'\beta$.
- Let's define the **best linear predictor** (the linear projection of Y_i on X_i):

$$\mathbb{E}^*[Y_i|X_i] = X_i'\beta_{OLS}$$

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Theorem

Suppose the CEF is linear. Then

$$\mathbb{E}[Y_i|X_i] = \mathbb{E}^*[Y_i|X_i]$$

OLS vs. CEF

- OLS works well when the CEF is linear: the two coincide.
- And when the CEF is *not* linear, OLS still gives the best linear approximation to it.

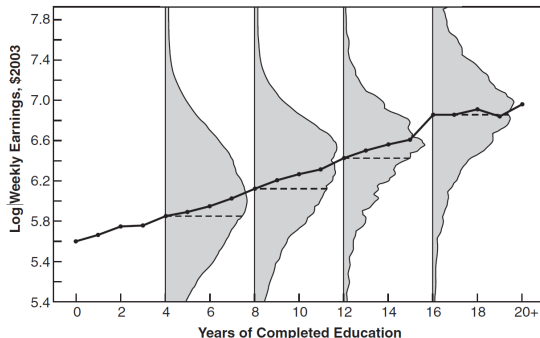


Figure 1: The OLS line and the CEF.

Interpretation of regressions

- Empirical papers until roughly the 1990s (probabilistic causality):
 - ▶ The regression was motivated by a complex model, and the equation was meant to describe an economic process.
 - ▶ All regressors on the right-hand side were more or less equally important.

Interpretation of regressions

- Empirical papers until roughly the 1990s (probabilistic causality):
 - ▶ The regression was motivated by a complex model, and the equation was meant to describe an economic process.
 - ▶ All regressors on the right-hand side were more or less equally important.
- Empirical papers nowadays (counterfactual causality):
 - ▶ Focusing on one causal relationship.
 - ▶ Identification strategy meant to eliminate selection bias.
 - ▶ Regressors are not equally important, and only one variable is seen as having causal effects.
 - ▶ All others are controls included in service of the focused causal agenda.

Reading the regression results

- We are interested in the relationship between wages, gender, and race.
- We have the following variables (a binary coding is a strong simplification of the actual data):

$$\text{Gender}_i = \begin{cases} 1 & \text{if Female} \\ 0 & \text{if Male} \end{cases}$$

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- Let's estimate:

$$\widehat{\text{Wage}_i} = \alpha + \beta_1 \times \text{Gender}_i + \beta_2 \times \text{Race}_i + \beta_3 \times \text{Gender}_i \times \text{Race}_i + \varepsilon_i$$

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- We have the results:

$$\widehat{\text{Wage}}_i = 10 - 5 \times \text{Gender}_i + 3 \times \text{Race}_i - 2 \times \text{Gender}_i \times \text{Race}_i$$

- What is the interpretation?

Main effect and interaction term

Let's assume that we have the following regression:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

Main effect and interaction term

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$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 0] = \alpha$

Main effect and interaction term

Let's assume that we have the following regression:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 0] = \alpha$
- $\mathbb{E}[y_i | x_{1i} = 1, x_{2i} = 0] = \alpha + \beta_1$

Main effect and interaction term

Let's assume that we have the following regression:

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i$$

- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 0] = \alpha$
- $\mathbb{E}[y_i | x_{1i} = 1, x_{2i} = 0] = \alpha + \beta_1$
- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 1] = \alpha + \beta_2$

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- $\mathbb{E}[y_i | x_{1i} = 0, x_{2i} = 1] = \alpha + \beta_2$
- $\mathbb{E}[y_i | x_{1i} = 1, x_{2i} = 1] = \alpha + \beta_1 + \beta_2 + \beta_3$

Note: If we include an interaction term, we should also include both main effects.

The saturated model

If x_1 is categorical rather than binary, we can saturate the model in x_1 :

$$y_i = \alpha + \sum_{j=1}^k \beta_j d_{ji} + \delta x_{2i} + \sum_{j=1}^k \gamma_j d_{ji} \times x_{2i} + \varepsilon_i$$

where x_1 takes $k + 1$ values and $d_{1i}, \dots, d_{ki}$ are dummies for those values, with one category omitted as the reference.

- The saturated model imposes no functional form on x_1 : it fits the conditional mean in every cell exactly.

Topics we will not cover

But you should know that they exist

- Frisch-Waugh-Lovell theorem
- R^2
- Weighting regressions
- Weighted least squares (WLS)
- Constraining coefficients (CLS)
- Maximum likelihood estimation (MLE)
- Generalized method of moments (GMM)
- Logit and probit models
- Normalization of variables
- . . .

Questions?

Summary

- Causality requires a prior ordering of variables into causes and effects.
- With observational data, the observed difference in means is ATT plus selection bias.
- Our job is to find a setting in which the selection bias is zero.
- OLS estimates the best linear predictor of Y given X ; it recovers the CEF when the CEF is linear.
- Interactions: always include the main effects, and be careful when interpreting the coefficients.

Next lecture: Statistical inference and IV. We will discuss the properties of OLS and how to make inference about the population parameters.

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